Eyes on the Road: Building a Vision System for Smart Driving

Abstract about the project:

This project presents an advanced lane detection system designed to identify road lane boundaries, estimate lane curvature, and compute the vehicle's lateral offset using classical computer vision techniques. Developed using Python and OpenCV, the system processes real-world driving footage to generate a reliable and accurate representation of lane geometry, crucial for autonomous driving applications.

The pipeline includes camera calibration for image distortion, perspective transformation to a bird's-eye view, combined color and gradient thresholding for feature extraction, and sliding window polynomial fitting for lane tracking. The system converts pixel-based detections into real-world measurements to calculate the curvature of the road and the vehicle's position relative to the centre of the lane.